Cloudie

CSP650 · CYBERSECURITY

Secure cloud storage that reads how a file is built

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THE IDEA

Signatures know the past. Structure reads the file.

Antivirus signatures only catch files someone has already seen. Cloudie also checks **how a file is put together**, so a disguised file is caught even when no engine has a signature for it yet. Every file is then encrypted before it is stored.

THE PROBLEM

01 New malware has no signature yet
Signature scanners miss files nobody has reported.

02 Attackers disguise files
A macro inside a .docx, `invoice.pdf.exe`, a ZIP hidden in a picture.

03 Stored files can leak
Files kept unencrypted are exposed if storage is breached.

HOW IT WORKS

Every upload is scanned, judged and encrypted before it is stored

Upload
Type allow-list, 25 MB limit, rate limit

VirusTotal
70+ engines, primary verdict
FAIL CLOSED

Structure checks
PDF, Office, ZIP, CSV, images, file name
NO SIGNATURE

AI for PDFs
Random Forest, 32 features, explained
EXPLAINABLE

AES-256
New key + IV per file, key goes to you

Cloud storage
Ciphertext only, on Google Cloud

INNOVATION · STRUCTURE LAYER

What the file is built from, by file family

FAMILY	WHAT THE RULES CATCH	MITRE ATT&CK
PDF	Hidden keywords (`/J#61vaScript`), exploit code in scripts, embedded programs	T1027 · T1059.007
Office	Macro inside a macro-free .docx, DDE, remote template	T1059.005 · T1221 · T1559.002
ZIP	Zip slip (`../`), zip bomb, programs disguised as documents	T1036.007 · CWE-22
CSV	Cells that run commands (`=cmd|…`) or leak data	T1559.002 · CWE-1236
Images	A ZIP, program or script hidden after the picture ends	T1027.003
Every file	Real type vs. its name, double extension, right-to-left trick	T1036.008 · T1036.002

Bytes are only read, never run or extracted, with a 3-second budget per file.

VERDICT ENGINE

VirusTotal decides first

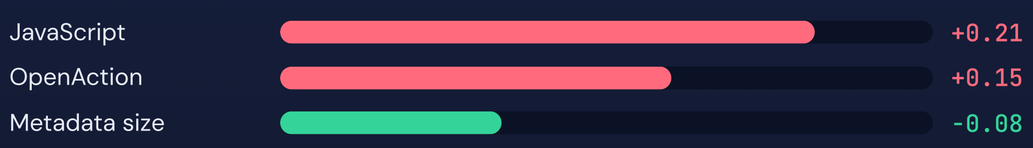
2 or more engines flag it	BLOCKED
No answer from VirusTotal	BLOCKED
Only 1 engine (classic false positive)	SUSPICIOUS
Clean, but a strong structure finding	SUSPICIOUS
Clean, but AI flags a risky PDF	SUSPICIOUS
Everything clean	CLEAN

Suspicious files are stored encrypted and tagged, with the reason shown in the scan report.

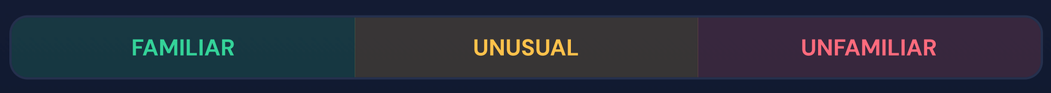
EXPLAINABLE AI

The report says why, and when it is unsure

Why this AI score: each feature's exact contribution from the trees.



Familiarity check: how close is this PDF to the PDFs it was trained on?



Bars show an example layout, not a measured file.

PROTECTING WHAT IS STORED

Only the owner holds the key

Key not stored
Only a SHA-256 hash of each file's key is kept

Integrity check
Decrypted bytes must match the hash taken at upload

Sealed audit log
HMAC hash chain: edits and deletions show up

2-step login
Password + one-time code, rate limited

Private scanner
Only the web service may call it (Cloud IAM)

QR key
Key delivered by email and QR, generated on our server

EVIDENCE · MEASURED, NOT ASSUMED

99.87%	Hold-out accuracy, 0 false positives
99.46%	Harder test: trained on small PDFs, tested on the largest
18,126	PDFs in CIC-Evasive-PDFMal2022
256	Automated tests (81 PHP + 175 Flask)

STANDARDS · MAPPED, NOT CERTIFIED

MITRE ATT&CK	12 techniques the checks detect
NIST CSF 2.0	Identify · Protect · Detect · Respond · Recover
ISO/IEC 27001	Annex A 8.5, 8.7, 8.13, 8.15, 8.24
OWASP	File Upload Cheat Sheet, Top 10
NIST 800-63B	Password and login rules

LIVE NOW · TRY IT YOURSELF

Scan, sign up, upload a file.

Runs on Google Cloud Run, Cloud SQL and Cloud Storage. For individuals, SMEs, schools, NGOs and government offices.

UN SDG 9 · Industry, Innovation & Infrastructure

UN SDG 16 · Strong Institutions

Live app

Source code